

Hariprakash Karthikeyan

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WORK EXPERIENCE

AI Engineer

March 2026 - Present

Community Dreams Foundation

Seattle, WA

- Architected a native knowledge base for Drema, DreamStream's multi-agent Gemini assistant (new specialist agent, async Cloud Tasks ingestion, hybrid pgvector + full-text retrieval), with SQL-level authorization so users only see permitted documents.
- Designed the service boundaries for a four-engineer build against a 15-requirement PRD and 9 acceptance criteria, separating ingestion pipeline and API/agent surfaces to avoid merge conflicts in a shared ~790-route Firebase Cloud Function.
- Designed, deployed, and own a production RAG service (LangChain, FAISS, OpenAI GPT-4o-mini, Streamlit) serving CDF volunteers organization-wide, with all answers grounded in official CDF policy and HR documents and cited to their exact source file.
- Tuned retrieval across 5 configs in two phases, validated via an 82-question automated eval suite re-run after every re-index that scores accuracy and grounding independently, improving end-to-end answer accuracy on the eval set from 45% to 89.2%.
- Built zero-downtime live document uploads and index rebuilds behind a password-protected admin UI, and hardened the service with input validation, metadata filtering, and full OpenAI API error handling across all failure modes for reliable uptime.

Data Scientist - Marketing

August 2025 - December 2025

Butler/Till

Rochester, NY

- Replaced an unstable DMA-level Media Mix Model ($R^2 = 0.55$) with a clustered hierarchical Bayesian model across 188 markets, improving validation R^2 by 13.7% and reducing budget allocation volatility driven by noisy local market-level estimates.
- Automated nonlinear adstock + Hill-based modeling workflows (PyMC + JAX) into a reusable pipeline, cutting scenario analysis time by 40–50% and increasing the number of budget simulations and scenarios evaluated in each quarterly planning cycle.
- Stabilized regional media effect estimates (cluster R^2 0.88–0.92) using partial pooling across market clusters, enabling consistent comparison of channel performance across regions and reducing contradictory cross-market budget recommendations.
- Modeled channel-specific carryover and saturation behavior (e.g., TV high carryover; Display early diminishing returns), supporting more disciplined incremental spend decisions and avoiding oversaturated channel allocations in future budget plans.

Data Science Intern

June 2025 - July 2025

Buffalo Solar

Buffalo, NY

- Engineered an automated monitoring pipeline consuming the SolarEdge REST API to process 15-minute telemetry from 468 solar sites, generating prioritized anomaly alerts and fully replacing the manual daily site-by-site performance review process.
- Shipped a Streamlit + Plotly dashboard adopted by engineering and ops teams, cutting manual monitoring by ~30–40%.
- Tuned anomaly detection precision to ~90% via residual analysis and threshold calibration, lowering false inspection flags.

Research Data Scientist

February 2025 - May 2025

University of Rochester Medical Center

Rochester, NY

- Built reproducible Python pipelines (Pandas, NumPy) to filter, clean, and merge 2,000+ longitudinal patient records across admissions tables, replacing manual chart review with reproducible, fully automated cohort construction for research.
- Integrated labs, admissions, and notes into time-aligned feature tables, giving researchers reusable modeling-ready data.
- Implemented regex-based NLP to turn unstructured clinical notes (e.g., LVEF values) into an 845-patient analysis-ready dataset.

PROJECTS

[Real-Time Music Data Platform & LLM Assistant | AWS S3, Kafka, PostgreSQL, LangChain, OpenAI, Python, Docker](#)

- Developed an event-driven platform processing 500K+ user events (~10GB), with idempotent consumers ensuring duplicate-free processing under retries and replays, and SQL aggregations in PostgreSQL feeding session-level analytics tables downstream.
- Partitioned S3 prefixes for raw events, sessions, and metadata, decoupling storage from compute for efficient queries.
- Integrated a SQL-grounded LLM assistant (LangChain, pgvector) with sub-2s end-to-end latency, combining vector retrieval with relational queries over the streaming data layer to serve personalized recommendations designed to improve session continuity.

[Distributed Sentiment Analytics | Databricks \(AWS\), Spark, Delta Lake, MLflow](#)

- Architected a Spark Structured Streaming medallion pipeline on Databricks ingesting 100K+ tweets with MLflow-packaged models served via Spark UDFs, using checkpointing and Delta Lake ACID transactions to guarantee exactly-once, fault-tolerant writes.

[Real-Time Stock Market Data Pipeline | Kafka, Snowflake, dbt, Airflow, Docker, Power BI](#)

- Built a containerized streaming ELT pipeline (Docker) moving live FintHub API quotes through Kafka producers and consumers into Snowflake, orchestrated by Airflow DAGs on a 1-minute schedule to keep downstream market data near real-time.
- Modeled bronze/silver/gold layers in dbt on Snowflake, transforming raw quote data into clean OHLC and KPI tables that power a live Power BI market dashboard, keeping the ingestion, transformation, and reporting layers fully decoupled from each other.

SKILLS

- **Programming & Query Languages:** Python, SQL, Data Structures & Algorithms, Object-Oriented Design
- **Libraries & Frameworks:** Pandas, NumPy, scikit-learn, TensorFlow, PyMC, JAX, LangChain
- **Visualization & Apps:** Plotly, Matplotlib, Seaborn, Streamlit, FastAPI, REST APIs
- **Statistical Methods:** Bayesian Modeling, Time-Series Forecasting, A/B Testing, Causal Inference, Predictive Modeling
- **Cloud & Data Platforms:** AWS (S3, EC2, ECR), PostgreSQL, NoSQL, Git, Linux
- **Big Data & Streaming:** Spark (PySpark, Structured Streaming), Kafka, Databricks, Delta Lake, Dask
- **NLP & LLM:** Transformers, Embeddings, RAG, pgvector, FAISS, Regex NLP
- **MLOps & Versioning:** MLflow, DVC, Docker, GitHub Actions (CI/CD)

EDUCATION

University of Rochester

Rochester, NY

Master of Science in Data Science

August 2024 - December 2025

SRM Institute of Science and Technology (SRMIST)

Chennai, India

Bachelor of Technology in Electronics and Communication Engineering

August 2020 - May 2024